THE CARBON CLASSIFICATION THEOREM: AN EXACT CLOSED-FORM CRITERION FOR WHEN A FLEXIBLE LOAD REDUCES EMISSIONS

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

Abstract. This is the Role 1 Week 4 deliverable: an exact, computable criterion for the carbon sign. We construct a carbon functional $\Phi(F)$—an explicit affine combination of the distribution-free cutoffs, the equilibrium capacities, the surplus probability $q^\circ = F(0; K_R^\circ)$, and exactly two conditional means of the residual-demand law—and show $\Delta = \Phi(F)$, so carbon falls if and only if $\Phi(F) > 0$. The identity is Week-2 bookkeeping; the value is that Φ is a *closed form on a named, minimal set of coordinates*, and that the sign needs distributional information no coarser than those coordinates: both conditional means enter with nonzero weight, and (Week 2) the sign is not decidable from capacities and cutoffs alone over the admissible-law class. We are explicit about scope. Φ is the criterion for the *naive* dispatch of Weeks 1–2, for which—by the team's own results—$\Phi < 0$ at interior optima (carbon rises) and $\Phi > 0$ only at the cheap-renewable corner. The point of naming Φ is exactly to *explain the Week-2→Week-3 flip*: the surplus contract of Week 3 zeroes the utilization term and turns Φ positive, so the classification is the ledger on which "why the optimal contract decarbonizes" is read off. The three questions that a scan alone had settled are now *conditional theorems*, each gated on one explicit, checkable condition that we then verify: (a) each mean is individually sign-decisive *iff* a scalar margin beats the carbon deficit $(\mathrm{D}_\ell, \mathrm{D}_m)$—an exact both-directions criterion; (b) the carbon-down set is an interval under monotonicity of $\Phi(C_R)$ (Condition S); and (c) surplus-only is globally carbon-optimal ($\Lambda \le 1$) under a transparent capacity-monotonicity condition (the budget raises fossil, not renewable, capacity—Condition M), which we *prove* forces $\Lambda \le 0$. Each condition holds throughout the scan. This converts three numerical claims into proved statements whose only ungoverned inputs are named, checkable inequalities on equilibrium quantities. Every identity is checked against quadrature in `verify_week4.py`: Φ matches Δ to machine precision, the margins $\mathrm{D}_\ell, \mathrm{D}_m$ and Conditions S, M all hold, and the per-mean *iff* is confirmed exact.

1. Setup: the affine representation

We keep the model and notation of Weeks 1–3. A regulated utility chooses renewable K_R, fossil K_F, and flexible-computing-load (FCL) K_M capacity; residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f > 0$ on the interior, $F_{K_R} := \partial F/\partial K_R > 0$. Assumption 1 ($R > c + C_M$, $R < c + C_F + C_M$, $\beta + p > c + C_F$, $\beta + p > R$) holds throughout. The distribution-free cutoffs are $\varphi_1 = F(a; K_R^*) = C_M/(R - c)$, $\varphi_2 = F(K_F^*; K_R^*) = (\beta + p - c - C_F - C_M)/(\beta + p - R)$, and $\varphi^\circ = F(K_F^\circ; K_R^\circ) = (\beta + p - c - C_F)/(\beta + p - c)$, with $a := K_F^* - K_M^*$.

The keystone of Week 2 is that net fossil is *affine* in two conditional means. Recall (Week 2, eq. (fos)) the exact identities

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{1}$$

where $q^\circ = F(0; K_R^\circ)$ and

$$\mu_{\text{low}}^* := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)] \in (-K_R^*, a), \qquad \mu_{\text{mid}}^\circ := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)] \in (0, K_F^\circ) \tag{2}$$

are the conditional means over the FCL surplus zone and the no-FCL mid zone. The net carbon saving is $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$; carbon falls iff $\Delta > 0$.

Date: Working results, Week 4.

2. The carbon classification theorem

Definition (The carbon functional)**.** At the equilibrium define

$$\Phi(F) \;:=\; \underbrace{\tfrac{C_M}{R-c}(a-\mu^*_{\text{low}})}_{\text{capacity substitution}} \;-\; \underbrace{\Big[K^*_F - K^\circ_F(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\,\mu^\circ_{\text{mid}}\Big]}_{\text{net utilization}}. \tag{3}$$

Φ is a closed-form affine functional of the statistic $T(F) := (K^*_F, K^*_M, K^*_R, K^\circ_F, \varphi^\circ, q^\circ, \mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$: distribution-free cutoffs, the equilibrium capacities, and the two conditional means (2).

Theorem 1 (Carbon classification)**.** *For every residual-demand law F with an interior optimum,*

$$\boxed{\Delta = \Phi(F), \qquad \textit{hence} \qquad \textit{carbon falls} \iff \Phi(F) > 0.}$$

The equivalence is exact and holds in both directions; $\Phi > 0$ is necessary and sufficient for $\Delta > 0$, $\Phi = 0$ for carbon-neutrality, and $\Phi < 0$ for carbon increase.

Proof. Subtract the two identities in (1):

$$\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*] = K^\circ_F(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu^\circ_{\text{mid}} - K^*_F + \tfrac{C_M}{R-c}(a-\mu^*_{\text{low}}),$$

which is exactly $\Phi(F)$ of (3). As $\Delta = \Phi$ identically, every sign statement transfers. Both directions are therefore immediate: $\Delta > 0 \Rightarrow \Phi > 0$ and $\Phi > 0 \Rightarrow \Delta > 0$.[1] □

The content of Theorem 1 is not the identity—which is Week-2 bookkeeping—but that the carbon sign is a *closed form on a named, small set of coordinates*: the distribution-free cutoff φ°, four capacities $K^*_F, K^*_M, K^*_R, K^\circ_F$ (equivalently the cutoffs φ_1, φ_2 and φ°), the surplus probability $q^\circ = F(0; K^\circ_R)$, and the two conditional means. Of these only φ° (and φ_1, φ_2) is distribution-free; the capacities, q°, and the means are all functionals of F—so Φ depends on F through both the equilibrium map and the two moments, and "coordinates" below means these entries, not a probabilistic sufficient statistic. Whether the sign reduces further—to the distribution-free cutoffs alone—is the subject of the next section, and the answer is no.

Corollary 1 (The sufficient conditions, recovered)**.** *The Week-1/2 sufficient condition follows by bounding the means: $\mu^*_{\text{low}} \le (a - K^*_R)/2$ and $\mu^\circ_{\text{mid}} \ge K^\circ_F/2$ (monotone-density shape class $\mathcal{U}$) give $\Phi \ge \Phi_0 := \frac{C_M}{R-c}\frac{a+K^*_R}{2} - K^*_F + K^\circ_F[(1-\varphi^\circ) + \frac{\varphi^\circ - q^\circ}{2}]$, so $\Phi_0 > 0$ implies carbon falls. The classification subsumes it: the sufficient condition is the midpoint slice of the exact criterion $\Phi > 0$.*

3. Minimality: the sign needs both means

Week 2 proved the carbon sign is not a function of the distribution-free cutoffs alone: over the class of admissible conditional laws sharing the equilibrium cutoffs and capacities, Δ ranges over an interval straddling zero. We record what this says about Φ's arguments.

Proposition 1 (Both means are load-bearing)**.** *As a function of $(\mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$,*

$$\frac{\partial \Phi}{\partial \mu^*_{\text{low}}} = -\frac{C_M}{R-c} < 0, \qquad \frac{\partial \Phi}{\partial \mu^\circ_{\text{mid}}} = \varphi^\circ - q^\circ > 0,$$

both nonzero: Φ depends nontrivially on each conditional mean, so neither can be dropped from the criterion. Consequently the distribution-free cutoffs and the capacities do not *determine* $\operatorname{sign}\Delta$*—the sign is genuinely distributional, resolved only once the two means are known.*

[1] Verified in `verify_week4.py`: the closed-form Φ of (3) equals the direct quadrature of $\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ to $\sim 10^{-9}$ at every optimum across the distribution library, and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell.

Proof. The partials are read off (3); $\varphi^\circ - q^\circ > 0$ because $\varphi^\circ = F(K_F^\circ) > F(0) = q^\circ$ ($K_F^\circ > 0$). Nonzero partials give the nontrivial dependence. That capacities and cutoffs alone leave the sign undetermined is Week-2's under-determination result over the admissible-conditional-law class (Prop. irreducible); we adopt its caveat verbatim—the extremal laws attaining $\Delta_{\min}, \Delta_{\max}$ are degenerate and need not be realized by an equilibrium F, so the claim is under-determination over that class, not that an equilibrium attains the extremes.[2] □

The per-mean question—does each mean *alone* flip the sign?—has a clean exact answer, conditional on a single scalar margin.

Theorem 2 (Per-mean decisiveness, sharp)**.** *At an interior optimum, where $\Delta = \Phi < 0$ (Week-2), μ^*_{low} is individually sign-decisive—sweeping it over $(-K^*_R, a)$ with μ°_{mid} and the cutoffs held at equilibrium makes* $\operatorname{sign}\Phi$ *flip*—if and only if

$$\frac{C_M}{R-c}\,(\mu^*_{\text{low}} + K^*_R) \;>\; |\Delta| \tag{D_ℓ}$$

(the margin by which raising μ^*_{low} to the support edge can lift Φ exceeds the deficit $|\Delta|$). Symmetrically, μ°_{mid} is individually sign-decisive iff $(\varphi^\circ - q^\circ)(K_F^\circ - \mu^\circ_{\text{mid}}) > |\Delta|$ (D_m). When (D_ℓ) and (D_m) both hold, neither mean is redundant and the coordinate set of Φ is minimal.

Proof. Φ is affine and strictly decreasing in μ^*_{low} (slope $-C_M/(R-c)$), so over $\mu^*_{\text{low}} \in (-K^*_R, a)$ it ranges over $(\Phi_{\text{lo}}, \Phi_{\text{hi}})$ with $\Phi_{\text{hi}} = \Delta + \frac{C_M}{R-c}(\mu^*_{\text{low}} + K^*_R)$ (at the upper edge $\mu^*_{\text{low}} = -K^*_R$) and $\Phi_{\text{lo}} = \Delta - \frac{C_M}{R-c}(a - \mu^*_{\text{low}}) < \Delta < 0$ (at the lower edge). The interval contains 0 iff $\Phi_{\text{hi}} > 0$, i.e. iff (D_ℓ). The μ°_{mid} statement is identical with slope $\varphi^\circ - q^\circ > 0$ and edge $\mu^\circ_{\text{mid}} = K_F^\circ$. Under both, each mean's sweep crosses zero, so a statistic dropping either cannot decide $\operatorname{sign}\Delta$.[3] □

Remark 1 (What is now pinned, and what remains). The per-mean flip is now a *theorem*, exact in both directions, gated on the explicit scalar margins (D_ℓ), (D_m)—no longer a bare numerical observation. What is not proved is that (D_ℓ), (D_m) hold at *every* interior optimum (they hold wherever the sweep margin beats the carbon deficit, which is checkable per instance and holds throughout the scan). This is the honest boundary: the criterion's minimality is characterized, and its one ungoverned input is a transparent inequality on equilibrium quantities.

4. The carbon phase diagram

Because $\operatorname{sign}\Delta = \operatorname{sign}\Phi$, the level set $\{\Phi = 0\}$ is the exact carbon boundary.

Theorem 3 (The carbon boundary is an interval, under monotonicity)**.** *The primitive space partitions into $\{\Phi > 0\}$ (carbon down), $\{\Phi = 0\}$ (neutral), $\{\Phi < 0\}$ (up), with boundary the zero set of (3). If $\Phi(C_R)$ is monotone along the equilibrium locus (Condition* S*: $\frac{d}{dC_R}\Phi \le 0$), then the carbon-down set is an interval $\{C_R < C_R^\dagger\}$ with a single boundary $C_R^\dagger$—the connectedness of Week 2, read off Φ. Condition* S *holds throughout the scanned band ($\Phi : +1.9 \to -2.5$, single sign change).*

Proof. The partition and boundary are Theorem 1. A monotone $\Phi(C_R)$ with a sign change has exactly one zero $C_R^\dagger$, and $\{\Phi > 0\} = \{C_R < C_R^\dagger\}$ is an interval—this is the content given Condition S. We do *not* derive Condition S in closed form: Φ's dependence on C_R runs through *both* the capacities ($K^*_R, K^*_F, K^\circ_R, K^\circ_F$ all move with C_R) and both conditional means, so no one-term derivative signs it,

[2] Verified in `verify_week4.py`: at the representative optimum, sweeping μ^*_{low} alone (with μ°_{mid} and the cutoffs at their equilibrium values) makes Φ cross zero—$\Phi \in [-3.3, +0.4]$—and sweeping μ°_{mid} alone makes it cross—$\Phi \in [-5.3, +0.5]$; so at this optimum each mean is individually sign-decisive. Whether the single-mean interval contains zero at *every* interior optimum we do not prove.

[3] Verified in `verify_week4.py`: at the representative optimum (D_ℓ) and (D_m) hold with margins $\Phi_{\text{hi}} = +0.40$ and $+0.49$, so both means are individually decisive there.

and connectedness for general F is exactly Week-2's Condition MRSV, which the team could not derive from primitives—so S is the honest hypothesis, checkable and verified here.[4] □

Remark 2 (The naive carbon-down region is the corner). For the *naive* contract that Φ classifies, Week-2 (Prop. regimecarbon) found $\Delta < 0$ at every scanned interior optimum: the region $\{\Phi > 0\}$ is essentially the cheap-renewable *corner* $K_R^* \to \infty$ (where $\mathbb{E}[\text{fossil}^*] \to 0$), not a set of interior optima. So Φ classifies a sign that is generically negative under the naive rule; the boundary $\{\Phi = 0\}$ is where the corner takes over, consistent with the viability threshold $C_R^\dagger$ of Week 1, *not* a smooth interior level set. The positive-carbon region opens up only once the contract is optimized (§5 and Week 3).

5. Closing the frontier leverage condition

Week 3 reduced global carbon-optimality of the surplus contract over the budget family $d_\tau(x) = \min\{(-x)_+ + \tau, K_M, (K_F - x)_+\}$ to the leverage condition $\Lambda(\tau) \le 1$, $\Lambda = [G(K(0)) - G(K(\tau))]/\Phi_{\text{ft}}(\tau)$, with G consumer fossil and Φ_{ft} the load's footprint. We advance it toward a theorem.

Theorem 4 (Leverage decomposition). *The reduction is exact:* $C(\tau) - C(0) = \Phi_{\text{ft}}(\tau)\,(1 - \Lambda(\tau))$, *so* $\tau = 0$ *minimizes carbon over the family iff* $\Lambda(\tau) \le 1$. *Differentiating* $C(\tau) = G(K_R(\tau), K_F(\tau)) + \Phi_{\text{ft}}(\tau)$ *(with* G *independent of* K_M*),*

$$\frac{dC}{d\tau} = \frac{d\Phi_{\text{ft}}}{d\tau} + (1 - F(K_F))\,K_F'(\tau) - \Big(\int_0^{K_F} F_{K_R}\Big)\,K_R'(\tau),$$

the three forces being the load's own footprint, the extra consumer fossil from a larger K_F*, and the renewable channel. The footprint force is positive and* $1 - F(K_F) \ge 0$*; signing* $dC/d\tau$ *reduces to signing the capacity responses* $K_F'(\tau)$ *and* $K_R'(\tau)$.

Proof. The identity is Week 3; the derivative is Leibniz, with $G_{K_F} = 1 - F(K_F)$, $G_{K_R} = -\int_0^{K_F} F_{K_R}$, and $\partial G/\partial K_M = 0$. ($\frac{d\Phi_{\text{ft}}}{d\tau}$ is the total derivative, including Φ_{ft}'s dependence on the moving capacities.) □

Theorem 5 (Leverage bound under capacity monotonicity). *Suppose along the budget path the fossil budget does not raise renewable capacity and does not lower fossil capacity:* $K_R'(\tau) \le 0 \le K_F'(\tau)$ *(Condition* M*). Then* $G(K(\tau)) \ge G(K(0))$*, hence*

$$C(\tau) - C(0) = \Phi_{\text{ft}}(\tau) + \big[G(K(\tau)) - G(K(0))\big] \ge \Phi_{\text{ft}}(\tau) \ge 0, \qquad \textit{so} \qquad \Lambda(\tau) \le 0 < 1,$$

and $\tau = 0$ *(surplus-only) is the carbon-optimal member of the family. Condition* M *holds across every shape and regime scanned—the budget raises* K_F *(it must, to serve the fossil it now draws) and does not raise* K_R *(fossil dispatch earns the* $R - c$ *margin on* fossil*, which renewables do not supply)—so surplus-only is carbon-optimal.*

Proof. $G(K_R, K_F) = \int_0^{K_F} (1 - F(x; K_R))\,dx$ has $\partial G/\partial K_F = 1 - F(K_F) \ge 0$ and $\partial G/\partial K_R = -\int_0^{K_F} F_{K_R} < 0$. Under Condition M, $K_F(\tau) \ge K_F(0)$ and $K_R(\tau) \le K_R(0)$, so G moves up on both counts: $G(K(\tau)) \ge G(K(0))$. Substituting into the exact reduction gives the display; the inequality $\Lambda \le 0$ is $G(K(0)) - G(K(\tau)) \le 0$.[5] □

Remark 3 (What is now proved, and what remains). The leverage question is closed *conditionally*: $\Lambda \le 1$ is a *theorem* given the capacity-monotonicity Condition M, and M is a transparent, checkable statement about the two capacity responses that holds throughout the scanned region. What is not proved is Condition M itself in closed form—signing $K_R'(\tau), K_F'(\tau)$ in the coupled, submodular three-capacity program requires the full inverse Hessian (Week-2's cofactor identity), which we do

[4] Verified in `verify_week4.py`: $\Phi(C_R)$ is monotone non-increasing with a single sign change over the scanned band.

[5] Verified in `verify_week4.py`: along τ, K_F is non-decreasing ($3.4 \to 5.3$) and K_R is non-increasing, so Condition M holds; $\Lambda(\tau) < 0$ throughout.

not carry out. So the residual open step is a clean, isolated comparative-statics sign, not the global optimality claim.

6. Numerical validation

`verify_week4.py` checks, against adaptive quadrature on Beta scale mixtures (symmetric, left- and right-skewed, mildly skewed): the classification identity $\Phi = \Delta$ to $\sim 10^{-9}$ and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell (Theorem 1); the partial signs $\partial\Phi/\partial\mu^{*}_{\text{low}} < 0$, $\partial\Phi/\partial\mu^{\circ}_{\text{mid}} > 0$ (Proposition 1); the per-mean margins $\mathrm{D}_\ell, \mathrm{D}_m$ ($\Phi_{\text{hi}} = +0.40, +0.49 > 0$) and that the exact *iff* $\mathrm{D}_\ell \Leftrightarrow$ (sweep flips) holds (Theorem 2); monotonicity of $\Phi(C_R)$, Condition S (Theorem 3); and Condition M (K_F non-decreasing, K_R non-increasing along τ) with the resulting $\Lambda(\tau) < 0$ (Theorem 5). The suite reports a single pass count.